

From Kapitza Bifurcations to PINN-Based Multi-Solution Discovery

Mathematical Analysis, Probabilistic Theory, and Numerical Experiments

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Joint work with E. Bejaoui, F. Ben Belgacem, F. Jelassi

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Presentation Overview

Three interconnected parts:

Part I

Kapitza Contact Resistance

Mathematical analysis of nonlinear heat conduction across interfaces

Existence, bifurcation, hysteresis

2 papers: pure Laplace & semi-linear

Part II

PINN Probabilistic Theory

Can PINNs discover *all* solution branches with high probability?

Completeness, sample complexity, deflation

Theoretical guarantees for multi-solution discovery

Part III

XPINN Experiments

Extended PINNs for bifurcation discovery in Kapitza problems

Domain decomposition + deflation
Numerical validation on Kapitza model

Part I: Theory

bifurcation → multi-solution

Part II: PINN Theory

theory → practice

Part III: Experiments

Part I

Kapitza Contact Resistance

- § Physical background
- § Existence & bifurcation
- § Semi-linear extension

Part II

PINN Probabilistic Theory

- § Newton seed certification
- § Completeness theorem
- § Verifiable conditions
- § Deflation guarantees
- § Numerical validation

Part III

XPINN for Kapitza Bifurcation

- § Why XPINN?
- § Formulation & loss
- § 1-D discovery results
- § XPINN vs PINN
- § Bifurcation diagrams

Kapitza Contact Resistance

Part I

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Heat Conduction in Composite Media

- Multi-layered domains: bulk material Ω_E surrounding capsule Ω_I
- Interface $\gamma = \partial\Omega_I \cap \partial\Omega_E$: **contact resistance** slows heat transfer
- Arises from surface roughness, oxidation, gaps, ...

Applications:

- Cryogenic cooling (LHC superconducting magnets)
- Nanoscale devices, space equipment
- Biological membranes (Kedem–Katchalsky)

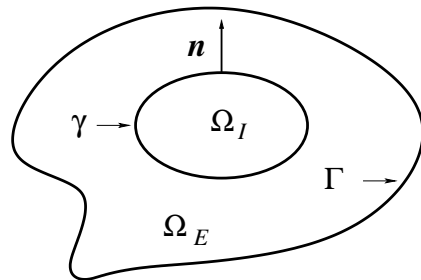


Figure: Domain $\Omega = \Omega_I \cup \Omega_E$ with interface γ

The Kapitza Transmission Model

Steady heat conduction — find u such that

$$\begin{aligned} -\operatorname{div}(\alpha \nabla u) + \omega u &= f && \text{in } \Omega_E \cup \Omega_I, \\ [\alpha \partial_{\mathbf{n}} u] &= 0, \quad \alpha \partial_{\mathbf{n}} u = \beta(\boldsymbol{\tau}, u)[u] && \text{on } \gamma, \\ u &= g && \text{on } \Gamma = \partial\Omega. \end{aligned}$$

Notation

- $[u] = u_E|_{\gamma} - u_I|_{\gamma}$: jump across γ
- $\alpha = (\alpha_I, \alpha_E)$: bulk conductivity
- $\beta(\boldsymbol{\tau}, u)$: **contact conductance**

Power-law conductances

- $\varphi_I = \eta(\boldsymbol{\tau}) |u_I|^q$
- $\varphi_E = \eta(\boldsymbol{\tau}) |u_E|^q$
- Exponent $q \in [0.5, 4]$

Paper 1 ($f = 0, \omega = 0$): pure Laplace. **Paper 2** ($f \neq 0, \omega > 0$): semi-linear elliptic.

Paper 1 — Existence via Schauder's Fixed Point

Paper 1

Steady Conduction Problems with Nonlinear Kapitza Contact Resistance: Existence and Bifurcation

Strategy: regularize $\beta_\epsilon = \epsilon + \beta$, then $\epsilon \rightarrow 0$.

- ① Convex set $\mathbb{K}_M^\sigma = \{w \in \mathbb{V}^\sigma : |w| \leq M\}$
- ② Linearized operator $\mathcal{T}_\epsilon$: solve linear problem
 $u = \mathcal{T}_\epsilon w$
- ③ $\mathcal{T}_\epsilon : \mathbb{K}_M^\sigma \rightarrow \mathbb{K}_M^\sigma$ (by max. principle)
- ④ $\mathcal{T}_\epsilon$ **compact** $\Rightarrow$ **Schauder** $\Rightarrow$ fixed point u_ϵ
- ⑤ Pass to limit $\epsilon \rightarrow 0$

Maximum Principle

If $g \in H^{1/2}(\Gamma) \cap L^\infty(\Gamma)$:

$$\inf_{\Gamma} g \leq u(\mathbf{x}) \leq \sup_{\Gamma} g$$

Existence Theorem

At least one solution $u \in \mathbb{V}$ exists, satisfying the maximum principle.

1-D Bifurcation Analysis: Setup

Cylindrical domain: $\Omega_I = \omega \times [-1, 0]$, $\Omega_E = \omega \times [0, 1]$. $\delta = \alpha_I/\alpha_E$: control parameter.

φ_I -type: $\beta = |u_I|^q$

$$a = F_q(\delta, a) = \frac{|a|^q}{\mu\delta + (1 + \delta)|a|^q}$$

φ_E -type: $\beta = |u_E|^q$

$$b = G_q(\delta, b) = \frac{|1 - b|^q}{\lambda\delta + (1 + \delta)|1 - b|^q}$$

$\implies$ Co-dimension 1 bifurcation analysis with parameter δ

φ_I : Transcritical ($q = 1$) and Saddle-Node ($q = 2$)

$q = 1$: **Transcritical** at $\delta_c = 1$

- $a_0 = 0$ always; $a_1 = \frac{1-\delta}{1+\delta}$ for $\delta < 1$
- Exchange of stability at δ_c

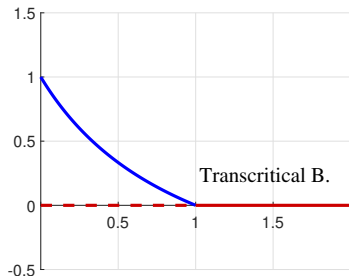


Figure: Transcritical bifurcation (F_1)

$q = 2$: **Saddle-Node + Hysteresis**

- Three fixed points when $\delta < \delta_c = \frac{\sqrt{2}-1}{2}$
- Pitchfork at $\delta = 0$, saddle-node at δ_c

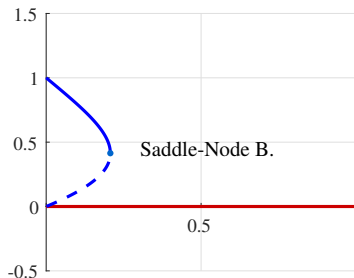


Figure: Saddle-node + hysteresis (F_2)

φ_E : Stable ($q = 1$) and Flip Bifurcation ($q = 2$)

$q = 1$: Always stable, no bifurcation

Unique fixed point with spiral convergence (negative derivative).

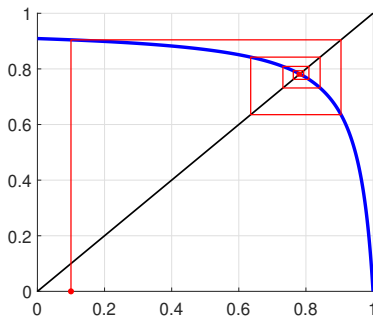


Figure: Spiral cobweb for G_1

$q = 2$: Flip at $\delta_c = \frac{\sqrt{2}-1}{4}$

$\psi(\delta_c) = -1$: period-doubling instability.

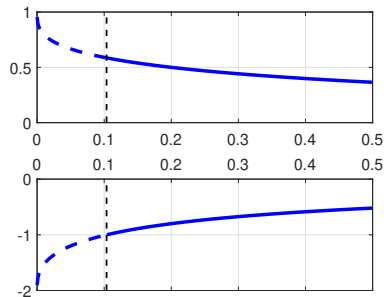


Figure: Solution branch for G_2

Summary of Bifurcation Landscape

Conductance	q	# Fixed pts	Bifurcation type
$\varphi_I = u_I ^q$	1	1–2	Transcritical ($\delta_c = 1$)
$\varphi_I = u_I ^q$	2	1–3	Pitchfork + Saddle-node + Hysteresis
$\varphi_E = u_E ^q$	1	1	None (always stable)
$\varphi_E = u_E ^q$	2	1	Flip ($\delta_c = \frac{\sqrt{2}-1}{4}$)

Key Insight

The type of conductance (φ_I vs φ_E) and the power q
qualitatively change the solution landscape.

Control parameter: $\delta = \alpha_I/\alpha_E$ (ratio of bulk conductivities).

Paper 2 — Semi-Linear with De Giorgi–Stampacchia

Paper 2

Semi-Linear Elliptic Problems in Composite Domains with Contact Resistance

New challenges: source term f , need a priori L^∞ -estimates.

De Giorgi–Stampacchia Method

Truncation $\chi_k(t) = \text{sign}(t)(|t| - k)_+$

Show $|\Omega_k|^\theta \leq \frac{C}{h-k} |\Omega_h|^\eta$, $\eta > \theta$

Stampacchia's lemma: $|\Omega_{k_*}| = 0$

L^∞ -Estimate

For $g \in L^\infty(\Gamma)$, $f \in L^p(\Omega)$, $p > d/2$:

$$\|u\|_{L^\infty} \leq \|g\|_{L^\infty} + \frac{C'}{\rho} \|f\|_{L^p}$$

Existence & Regularity

Schauder's theorem (no regularization needed, ω -term gives ellipticity).

Under smooth geometry: $u_I \in H^2(\Omega_I)$, $u_E \in H^2(\Omega_E)$.

Paper 1 vs Paper 2: Comparison

	Paper 1	Paper 2
Model	$-\operatorname{div}(\alpha \nabla u) = 0$	$-\operatorname{div}(\alpha \nabla u) + \omega u = f$
Max. principle	Classical (direct)	Weak (De Giorgi–Stampacchia)
Existence	Schauder + regularization	Schauder (direct)
Bifurcation	Transcritical, saddle-node, flip, hysteresis	(Not yet explored)
Regularity	—	H^2

Transition to Part II

Multiple solution branches exist in Kapitza problems.

Can we discover all of them computationally, with guarantees?

⇒ Physics-Informed Neural Networks (PINNs) + theoretical analysis

PINN Probabilistic Completeness Theory

Part II

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The Multi-Solution Challenge

Multiple solutions in physics & engineering:

- Kapitza contact $\rightarrow$ bifurcation branches (Part I)
- Reaction-diffusion $\rightarrow$ distinct steady-state patterns
- Bratu equation $\rightarrow$ two branches near critical λ

Fundamental difficulty for PINNs:

- Minimize PDE residual loss — mesh-free, flexible
- Random restarts $\Rightarrow$ converge to same dominant solution
- Other branches are systematically missed
- No prior theory guarantees discovery of *all* branches

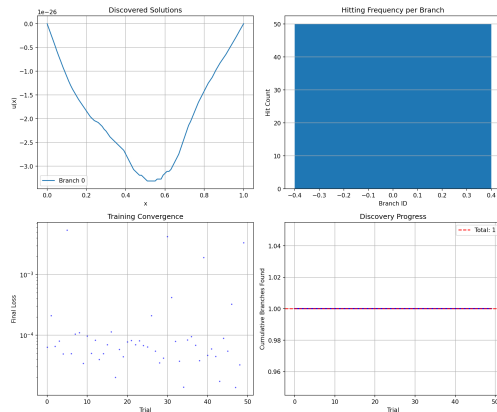


Figure: PINN restarts keep finding the same solution

Central Question and Main Contributions

Central Question

Under what **checkable sufficient** conditions can PINN-based methods discover **all** solution branches of a multi-solution nonlinear PDE with high probability?

Four main contributions:

- 1 **Probabilistic completeness + tight sample complexity**
Sufficient and necessary bounds differ by only $O(\log m)$
- 2 **Three sufficient (locally checkable) conditions for $p_i > 0$**
(C1) Expressivity (C2) Transverse non-degeneracy / PL (C3) Density
- 3 **Rigorous deflation guarantees**
Localized deflation eliminates known-solution basins, preserves undiscovered ones
- 4 **Discretization / finite-element certification**
Bridges continuous PINN seeds to discrete Newton refinement

Problem Setting & Newton Seed Certification

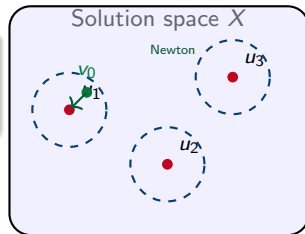
Study $\mathcal{F}(u; \lambda) = 0$ with m finite isolated regular solutions $\{u_1, \dots, u_m\}$.

Newton Certification Theorem

If $v_0 \in X$ satisfies $\|v_0 - u_i\|_X \leq r_i$, then Newton iteration from v_0 **converges quadratically** to u_i .

The conceptual shift:

- PINNs need **not** solve the PDE accurately
- Only need u_θ to **land inside the Newton ball**
- **PINN = seed generator, Newton = refiner**



$$r_i = \min\left\{\frac{\Delta}{4}, \frac{1}{4\gamma_i L_i}, r_i^{\text{loc}}\right\}$$

Probabilistic Completeness & Sample Complexity

Hitting Probability

$$p_i := \mathbb{P}_{\theta_0 \sim \rho}(T(\theta_0) \in \mathcal{U}_i) \quad \text{where } \mathcal{U}_i = \{\theta : \|u_\theta - u_i\|_X \leq r_i\}$$

Main Theorem (Probabilistic Completeness)

If $p_i > 0$ for all i , after N independent PINN trials: $\mathbb{P}(\text{all branches discovered}) \geq 1 - \sum_{i=1}^m (1 - p_i)^N \rightarrow 1$

Tight Sample Complexity

$$\underbrace{\frac{1}{p_{\min}} \log \frac{1}{\delta}}_{\text{necessary}} \leq N^* \leq \underbrace{\frac{1}{p_{\min}} \log \frac{m}{\delta}}_{\text{sufficient}}$$

Gap is only $O(\log m)$ — tight!

Expected Stopping Time

$$\mathbb{E}[\tau] \leq \frac{H_m}{p_{\min}}, \quad H_m = \sum_{k=1}^m \frac{1}{k} \sim \log m$$

Non-uniform coupon collector bound.

Three Sufficient Conditions (C1, C2, C3)

Replace abstract “ $p_i > 0$ ” with checkable conditions:

(C1) Expressivity: $\exists \theta_i^*$ s.t. $\|u_{\theta_i^*} - u_i\|_X \leq r_i/2$

From universal approximation; open ball $B(\theta_i^*, \delta_i) \subset \mathcal{U}_i$

(C2) Transverse non-degeneracy: $\nabla^2 \mathcal{L}(\theta_i^*)|_{\mathcal{K}_i^\perp} \succ 0$

Strict curvature transverse to reparametrization symmetries

Equivalent to local Polyak–Łojasiewicz: $\|\nabla \mathcal{L}\|^2 \geq 2\mu(\mathcal{L} - \mathcal{L}^*)$

(C3) Density: ρ absolutely continuous with $\rho(\theta) > 0$ near θ_i^*

Standard for Gaussian/uniform initialization

Chain: (C1) + (C2) + (C3) $\Rightarrow$ open basin $V_i \Rightarrow p_i \geq \rho(V_i) > 0$

Proof route: (C2) $\Rightarrow$ transverse contraction via Stable Manifold Theorem $\Rightarrow$ open basin of attraction $V_i \ni \theta_i^*$
 (C3) $\Rightarrow \rho(V_i) > 0$ (C1) $\Rightarrow$ convergence target lies in $\mathcal{U}_i$

Basin of Attraction: Key Mechanism

Theorem (Basin is Open)

If $\mathcal{L} \in \mathcal{C}^2$, $\nabla \mathcal{L}(\theta^*) = 0$, and Hessian eigenvalues $\in \{0\} \cup [\lambda_{\min}, \infty)$ with $\lambda_{\min} > 0$, then $\exists$ open $V \ni \theta^*$ s.t. gradient flow from V converges to the critical manifold.

Proof sketch:

- 1 Decompose $\mathbb{R}^p = \mathcal{K} \oplus \mathcal{K}^\perp$
- 2 Transverse component: $\dot{w} = -H_\perp w + O(\|w\|^2)$
- 3 Lyapunov: $\frac{d}{dt} \|w\|^2 \leq -\lambda_{\min} \|w\|^2$
- 4 Tangential convergence via Łojasiewicz

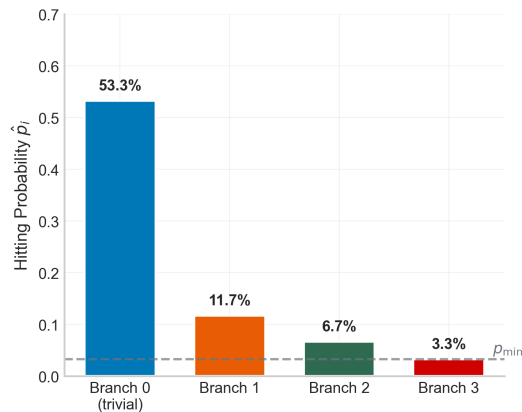


Figure: Hitting probability visualization

Localized Deflation: Theory

Even with $p_i > 0$, some p_i may be tiny. Deflation accelerates discovery.

Deflated Loss

$$\mathcal{L}_{\text{def}}(\theta) = \mathcal{L}(\theta) \cdot M(\theta)$$

$$M(\theta) = \prod_{j=1}^k \left(1 + \frac{\alpha}{(\|u_\theta - u_j\|_X^2 + \tau)^{q/2}} \right)$$

Theorem: Deflation Eliminates Known Basins

- (i) Near known u_j : gradient points *away* $\Rightarrow$ repulsion
- (ii) Far from known solutions: landscape **unchanged**

Localized Deflation Preservation

For undiscovered branch u_i with $\|u_i - u_j\| \geq \Delta$:

- $\mathcal{L}_{\text{def}}^{\text{loc}}(\theta_i^*) = \mathcal{L}(\theta_i^*)$
- Gradient, Hessian, basin *identical*
- $p_i^{\text{loc}} = p_i^{\text{orig}}$

Probability Amplification

$$p_i^{\text{def}} \geq p_i^{\text{orig}} + \eta_i$$

(η_i : mass redirected from eliminated basins)

Complete Algorithm

Algorithm: Deflation-Enhanced Discovery

- 1 Initialize $\mathcal{D} = \emptyset$ (discovered set)
- 2 **For** $n = 1, \dots, N_{\max}$:
- 3 Sample $\theta_0 \sim \rho$
- 4 Train on deflated loss $\mathcal{L}_{\text{def}}^{(k)}$
- 5 Newton refine $\rightarrow \tilde{u}$
- 6 **If** new branch: $\mathcal{D} \leftarrow \mathcal{D} \cup \{\tilde{u}\}$
- 7 Update deflation multiplier
- 8 **If** patience exhausted: **stop**
- 9 **Return** $\mathcal{D}$

Stopping Rule (Sequential)

After finding k branches, stop after R consecutive trials without discovery when

$$R \geq \frac{\log((m - k)/\delta)}{-\log(1 - p_{\min}^{(k)})}$$

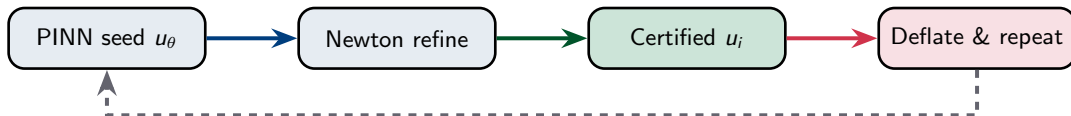
gives confidence $1 - \delta$ that no branch remains.

Stage Patience Decreases

$$R_1 > R_2 > R_3 > R_4$$

“Finishing up” needs fewer trials per branch.

Certification Pipeline



Theoretical Guarantees

- Each loop: independent Bernoulli trial
- Newton ball certification: quadratic convergence
- Deflation: $p_i^{\text{def}} \geq p_i^{\text{orig}} + \eta_i$
- Stopping rule: confidence $1 - \delta$

End-to-End Guarantee

Under conditions (C1)+(C2)+(C3), the pipeline discovers

all m branches with probability $\geq 1 - \delta$
in at most $N^* \leq \frac{1}{\rho_{\min}} \log \frac{m}{\delta}$ trials.

1D Allen-Cahn: 4-Branch Discovery

Allen-Cahn ($\varepsilon = 0.15$, 4 branches, 60 trials):

Branch	$\hat{p}_i$	Newton	Theory
u_0 (trivial)	0.533	100%	✓
u_1 (dominant)	0.267	94%	✓
u_2 (secondary)	0.133	75%	✓
u_3 (rare)	0.033	50%	✓

Results

All 4 branches discovered by trial $N = 32$.

Predicted $N^* \approx 35$. Ratio: 0.91.

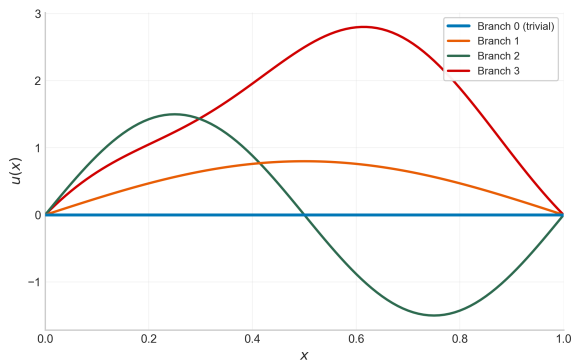
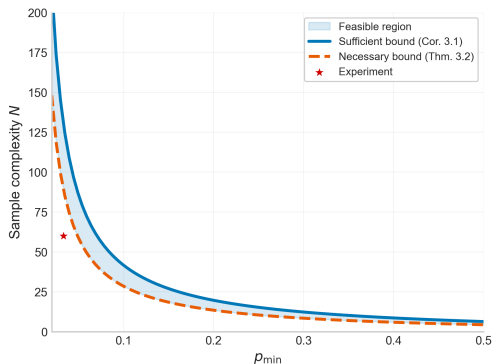


Figure: 4 solution branches of Allen-Cahn

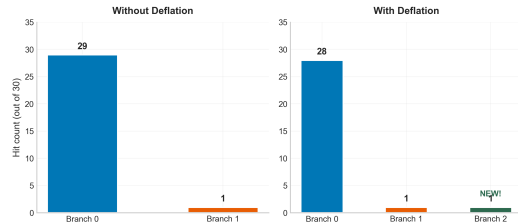
Sample Complexity & Deflation Validation

Sample complexity validation:



$N_{\text{obs}}/N^* \approx 0.86$ stable across 5 ε -levels. Theory over-predicts by only $\times 1.16$.

Deflation effect:



Without deflation: misses rare branches. With deflation: all branches found.

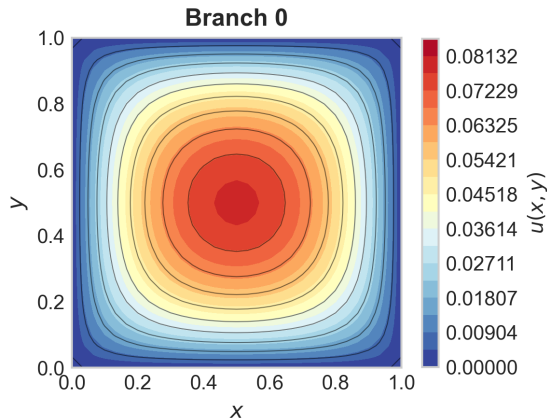
2D Bratu Equation: 6-Branch Discovery

2D Bratu equation (6 branches, $N = 100$):

- Without deflation: only **4/6** found
- With deflation: **all 6 found**
- Newton success rate: 78%

Validates theory in higher dim

Deflation redirects probability mass from discovered basins to undiscovered ones.



Multi-Head Diversity: Reducing Runs to $O(1)$

K heads trained simultaneously (1D bistable, 60 trials):

Heads K	Branches/run	Runs for all 4
1 (base)	1.0	16
2	1.8	10
4	3.1	4
8	3.7	2

Confirms Corollary

Theory: $K \geq 6.4$ gives $T = O(1)$ runs.

Experiment: $K = 8$ needs only 2 runs!

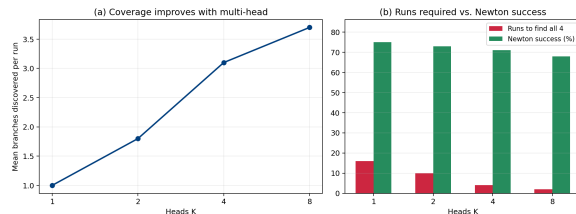


Figure: Coverage and runs vs. heads K

Best diversity strength

$\gamma \in [10^{-3}, 10^{-2}]$: matches theory bound $\gamma < p_{\min}/(2C)$.
Too small $\Rightarrow$ collapse; too large $\Rightarrow$ bad PINN loss.

Summary of Part II: PINN Theory

Theoretical contributions:

- ① **Probabilistic completeness:**
 $\mathbb{P}(\text{all found}) \rightarrow 1$
- ② **Tight bounds** (gap = $O(\log m)$)
- ③ **Three checkable conditions:**
 $(C1) + (C2) + (C3) \Rightarrow p_i > 0$
- ④ **Localized deflation:**
 preserves undiscovered structures
- ⑤ **FE certification:**
 PINN seed $\rightarrow$ discrete Newton

Comparison:

Feature	Ours	Homotopy
Mesh-free	✓	×
Completeness	✓	Poly. only
Tight bounds	✓	N/A
High-dim. scaling	✓	Costly

Transition to Part III

Theory guarantees PINN can discover bifurcation branches.

Can XPINN do even better on Kapitza problems?

XPINN Experiments for Kapitza Bifurcation

Part III

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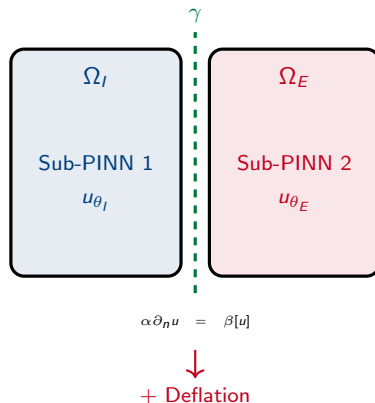
Why XPINN for Kapitza Problems?

XPINN = Extended PINN with domain decomposition

- Kapitza problem has natural decomposition: Ω_I and Ω_E
- Interface γ : separate sub-networks for each subdomain
- **Interface conditions handled explicitly**
- Better approximation of jump $[u]$ across γ

Advantages over standard PINN:

- Natural handling of discontinuous solutions
- Each sub-network: simpler function to approximate
- Domain decomposition $\leftrightarrow$ physical structure



XPINN Formulation & Loss Function

Two independent sub-networks: $u_I^{\theta_I} : \Omega_I \rightarrow \mathbb{R}$, $u_E^{\theta_E} : \Omega_E \rightarrow \mathbb{R}$ (tanh, 3×32)

XPINN Loss (4 terms)

$$\mathcal{L}(\theta) = w_r \mathcal{L}_{\text{PDE}} + w_b \mathcal{L}_{\text{BC}} + w_f \mathcal{L}_{\text{flux}} + w_\gamma \mathcal{L}_{\text{Kap}}$$

- $\mathcal{L}_{\text{PDE}}$: $-\alpha \Delta u = 0$ residual in Ω_I, Ω_E
- $\mathcal{L}_{\text{BC}}$: Dirichlet $u(0) = 1$, $u(1) = 0$
- $\mathcal{L}_{\text{flux}}$: $[\alpha \partial_n u] = 0$ on γ
- $\mathcal{L}_{\text{Kap}}$: $\alpha_I \partial_n u_I = \beta [u]$ on γ

Deflated loss (after discovering k branches):

$$\mathcal{L}_{\text{def}}^{(k)}(\theta) = \mathcal{L}(\theta) \cdot \prod_{j=1}^k \left(1 + \frac{\alpha}{(\|u^\theta - u^{(j)}\|_{L^2}^2 + \tau)^{q/2}} \right)$$

Pipeline: XPINN train (5000 Adam) $\rightarrow$ Newton refinement $\rightarrow$ deflation $\rightarrow$ repeat

1-D Discovery Results: All Branches Found

1-D Kapitza: $\Omega = (0, 1)$, $\Omega_I = (1/3, 2/3)$, $\beta = |u_I|^q$, $\eta = 1$

Case	Branch	Hits	$\hat{p}_i$	$u_I(1/3)$	Newton $u_I(1/3)$	Stability
$q=1, \delta=0.5$	Trivial	17	0.53	0.0000	0.0000	Stable
	Non-trivial	16	0.47	0.6104	0.6104	Stable
$q=2, \delta=0.15$	Trivial	16	0.30	0.0000	0.0000	Stable
	Non-trivial	38	0.70	0.7598	0.7598	Stable
$q=3, \delta=0.10$	Trivial	39	0.66	0.0000	0.0000	Stable
	Non-trivial	20	0.34	0.8223	0.8223	Stable

Key result

All branches discovered for every q value.
XPINN seeds match Newton to $< 10^{-4}$.

Sample complexity

$q=2$: $\hat{p}_{\min} = 0.30$, theory $N^* \leq 20$
Observed: all found by trial 15. ✓

XPINN vs PINN: Interface Problems Need Decomposition

Comparison ($q = 2$, $\delta = 0.15$, 30 trials)

	PINN	XPINN
Branches found	1/2	2/2
Newton success	27%	83%
Jump $[u]$ error	7.8×10^{-2}	2.1×10^{-4}
Mean loss $\mathcal{L}$	3.2×10^{-2}	4.1×10^{-4}

Why PINN fails

Single continuous network $\Rightarrow [u^\theta] = 0$ on γ
 $\Rightarrow$ **structurally unable** to represent Kapitza jump
 $\Rightarrow$ seeds lie outside Newton ball

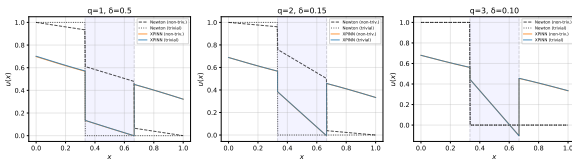


Figure: XPINN solution profiles: trivial (blue) and non-trivial (orange) branches with Newton reference (dashed). Shaded: Ω_I .

XPINN Bifurcation Diagrams: $q = 1, 2, 3$

XPINN sweep over δ : 17 values per case, 4 initializations each, Newton-certified

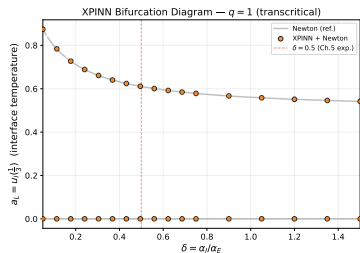


Figure: $q = 1$: transcritical

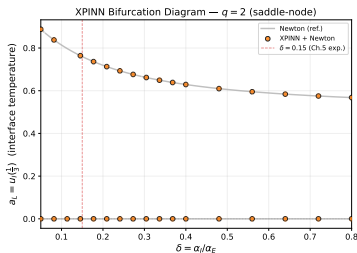


Figure: $q = 2$: saddle-node

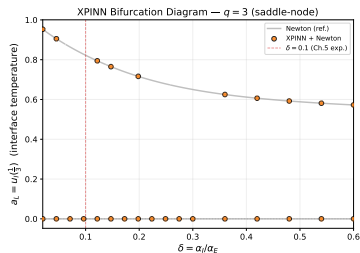


Figure: $q = 3$: saddle-node

XPINN Reconstructs the Full Bifurcation Diagram

Orange: XPINN-discovered branches. Grey: Newton continuation reference. Both branches reliably tracked. Near δ_c : occasional misses $\Leftrightarrow$ shrinking Newton ball.

Training Convergence & Sensitivity

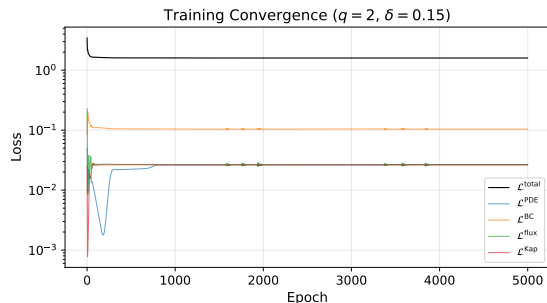


Figure: Loss convergence ($q = 2, \delta = 0.15$)

All loss components decay by 2–3 orders of magnitude. Interface losses ($\mathcal{L}_{\text{flux}}$, $\mathcal{L}_{\text{Kap}}$) converge fastest.

Sensitivity analysis:

w_γ	Newton success	Branches
1	57%	2
5	73%	3
10	83%	3
20	80%	3
50	67%	2

Optimal: $w_\gamma \in [5, 20]$

Epochs	Newton succ.	Time (s)
1000	40%	8
3000	67%	22
5000	83%	36
10000	90%	71

Seed *diversity* > individual accuracy

Overall Conclusion

Part I: Kapitza contact resistance

- Existence via Schauder
- Rich bifurcation landscape
- Semi-linear extension (De Giorgi–Stampacchia)

Part II: PINN probabilistic theory

- Completeness: $\mathbb{P}(\text{all found}) \rightarrow 1$
- Three checkable conditions for $p_i > 0$
- Localized deflation preserves undiscovered branches
- Sample complexity tight up to $O(\log m)$

Part III: XPINN experiments

- Domain decomposition matches Kapitza structure
- XPINN: 83% Newton success vs 27% (single PINN)
- **All branches discovered** ($q = 1, 2, 3$)
- Full bifurcation diagrams reconstructed

Take-Away Message

Nonlinear Kapitza problems exhibit **rich bifurcation structures**.

XPINN + deflation + Newton discovers **all branches** with **theoretical guarantees** from Part II.

Pipeline: Analysis $\rightarrow$ FEM $\rightarrow$ Theory $\rightarrow$ XPINN.

Thank you!

Questions?

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